ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Forms of energy, second-class levers, chain drives and torque; switch logic, power and resistance; indirect control of a cylinder with a memory valve and the retracting force; DC motors and stopping a motor from two places; safety colours, hazards and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw all symbols and circuits neatly, using a ruler.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

QUESTION 1

1.1 Name FOUR forms of energy. (4)

1.2 FIGURE 1 below shows a wheelbarrow, which is a lever.

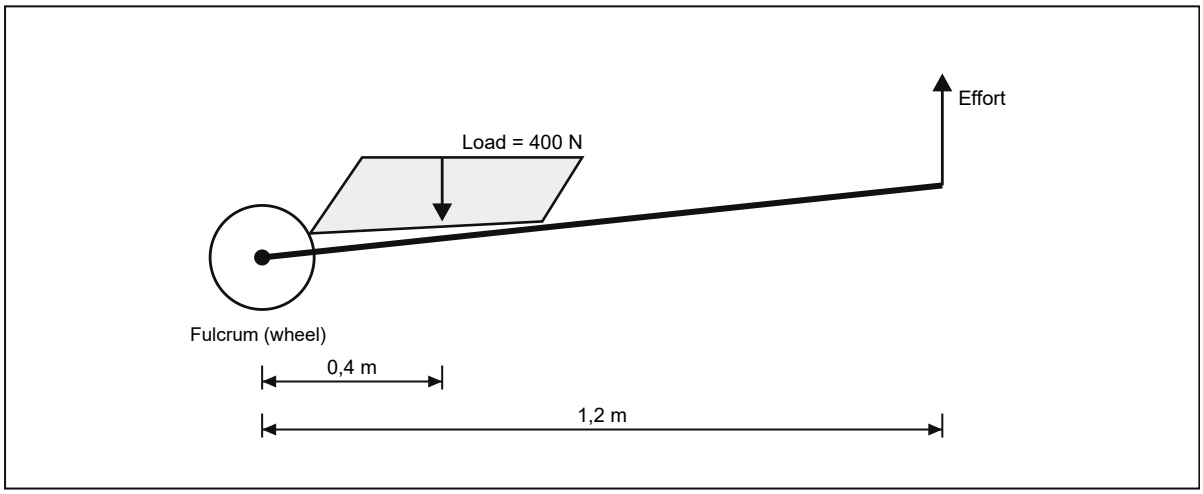


FIGURE 1

1.2.1 Identify the class of the lever. (1)

1.2.2 Calculate the effort needed to lift the load. (3)

1.2.3 Calculate the mechanical advantage. (2)

1.3 A driver sprocket with 18 teeth turns at 60 r/min and drives a sprocket with 54 teeth by means of a chain. Calculate the speed of the driven sprocket. (3)

1.4 State the function of a clutch. (2)

1.5 Define the term *torque* and state its SI unit. (3)

1.6 A force of 50 N is applied at the end of a spanner that is 0,3 m long. Calculate the torque. (2)

[20]

QUESTION 2

2.1 FIGURE 2 below shows a switching circuit.

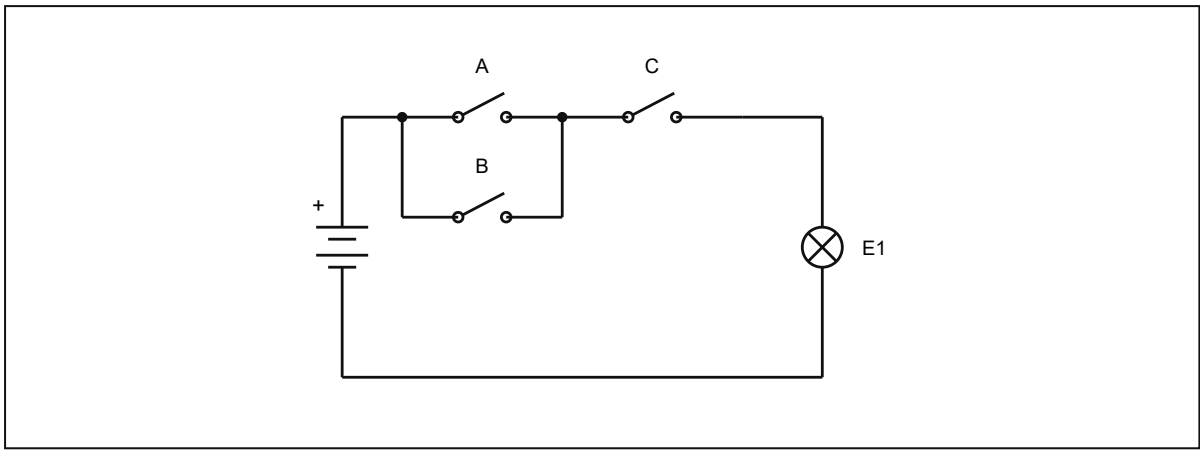


FIGURE 2

2.1.1 Name the logic function formed by switches A and B. (1)

2.1.2 Write the Boolean expression for lamp E1. (2)

2.1.3 Draw the truth table for the circuit. (4)

2.2 A 24 V lamp draws a current of 0,5 A. Calculate the following:

2.2.1 The power of the lamp (2)

2.2.2 The resistance of the lamp (2)

2.3 Explain the difference between a push button and a latching (toggle) switch. (2 × 2) (4)

2.4 Draw the symbol of a relay coil K1 and of a normally closed contact of K1. (3)

2.5 What is a multimeter used for? (2)

[20]

QUESTION 3

3.1 FIGURE 3 below shows the indirect control of a double-acting cylinder.

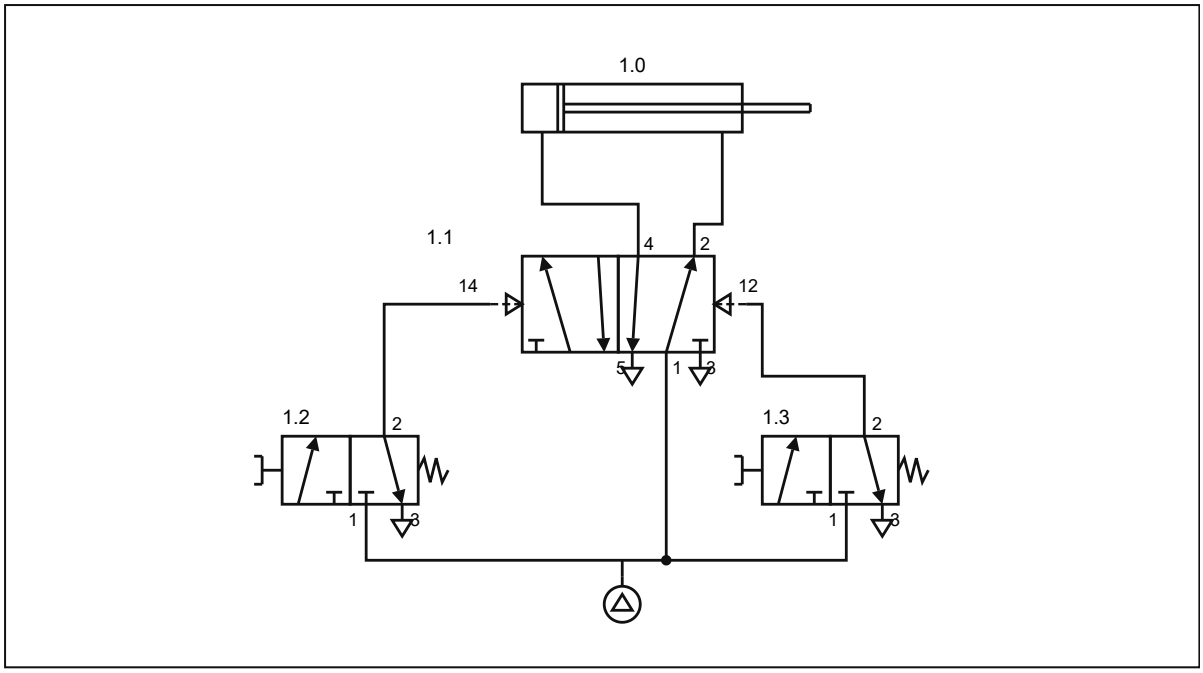


FIGURE 3

3.1.1 Name valve 1.1. (2)

3.1.2 Explain why valve 1.1 is called a memory valve. (3)

3.1.3 Describe what happens when valve 1.2 is pressed and released, and then valve 1.3 is pressed and released. (4)

3.2 Explain the difference between direct and indirect control of a cylinder. (2 × 2) (4)

3.3 A double-acting cylinder has a piston diameter of 40 mm and a piston-rod diameter of 16 mm. The air pressure is 600 kPa. Calculate the force when the cylinder retracts. (5)

3.4 What do the numbers 5 and 2 in the name "5/2-way valve" mean? (2)

[20]

QUESTION 4

4.1 Name FOUR main parts of a DC motor. (4)

4.2 Draw a control circuit in which a motor contactor K1 can be started from ONE place (START) and stopped from TWO places (STOP 1 and STOP 2). The circuit must have a holding contact and an overload contact (F1). (10)

4.3 Explain why an emergency stop button must have a normally closed (NC) contact. (2)

4.4 Explain the difference between alternating current (AC) and direct current (DC). (2 × 2) (4)

[20]

QUESTION 5

5.1 State the meaning of EACH of the following safety colours:

5.1.1 Red (1)

5.1.2 Yellow (1)

5.1.3	Blue	(1)
5.1.4	Green	(1)
5.2	List FOUR hazards in a mechatronics workshop.	(4)
5.3	Name FOUR items that should be in a workshop first-aid kit.	(4)
5.4	Explain the term *fault-finding* and name TWO fault-finding methods.	(4)
5.5	Give FOUR reasons why machines must be cleaned regularly.	(4)
		[20]
		TOTAL: 100

END OF QUESTION PAPER